

Study of honeybee learning through Reinforcement Learning models

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Abstract

This report presents the study of honeybee decision-making process through the application of reinforcement learning algorithms, especially Multi-Armed Bandit models. It details the methodology, encountered difficulties, and the progress made in the development of a web-based dashboard application for data visualization and analysis.

Keywords

Artificial Intelligence; Reinforcement Learning; Biology; Web Application;

Contents

I	Report	3
I.1	Introduction	3
I.2	Multi-Armed Bandit problem	3
I.3	Methodology and organization	3
I.4	Advancement	4
I.5	Difficulties	4
I.6	Conclusion	4
A	Bibliography	5

I.1 Introduction

Reinforcement learning is a subfield of Artificial Intelligence and Machine Learning allowing agents, such as humans, robots, or other living organisms, to learn optimal behavior by interacting with their environment. Nowadays, it is often combined with deep learning to develop autonomous programs capable of improving their performance over time through accumulated experience. To give an example, it has been very successful in domains such as chess engine programming, and more generally game theory, in which artificial intelligence algorithms already easily outperform humans in both strategy and tactics.

Beyond games, reinforcement learning also finds applications in biology where AI models can provide a better understanding of the decision-making process of living organisms. This project aims to explore Multi-Armed Bandit algorithm capacities in solving a sequential learning problem and compare their strategies with honeybees' responses to the same problem. The objective was to provide a web-based dashboard application that facilitates both data visualization and decision analysis.

I chose this subject since I have a particular interest in Artificial Intelligence and I love programming. In fact, AI offers opportunities for experimentation, particularly in domains such as computer vision, sports prediction, and strategic games like chess.

I.2 Multi-Armed Bandit problem

In probability theory and machine learning, the Multi-Armed Bandit (MAB) problem belongs to a family of algorithms in which an agent, placed in an experimental environment, must iteratively select one of multiple fixed choices, called arms or actions. At the beginning of the experiment, the agent has no prior knowledge of the outcome associated with each action. For instance, it may be rewarded or penalized and has to progressively learn which option is best by establishing a strategy, also called a policy. The agent's policy alternates between exploration and exploitation. During exploration, the agent takes a previously untested action, hoping it will lead to more rewards. On the other hand, the exploitation consists of the agent choosing the action giving the best rewards at the moment. The Multi-Armed Bandit is a Reinforcement Learning problem where the agent's primary objective is to maximize reward while minimizing regret.

For this project, bees (the agents) enter a Y-maze and land on one of two visual stimuli. Depending on its choice, each bee receives either a reward (sugar) or a punishment (a bitter substance). The bee then returns to its hive and repeats the process. During the experiment, data is collected and analyzed by researchers to understand learning and decision-making dynamics. In brief, bees' choices are compared with optimal solutions to the problem.

I.3 Methodology and organization

During this internship at IRIT (Institut de Recherche en Informatique de Toulouse), I was responsible for programming the application supporting the import, processing and analysis of collected data. I worked closely with my project supervisor, Ms. Emmanuelle Claeys, an engineer and researcher in artificial intelligence, as well as Ms. Elena Kerjean, a final-year PhD student at the Center for Integrative Biology (CBI). This project is part of her doctoral research. At the beginning of February, I first focused on studying the mathematical concepts of Reinforcement Learning, understanding the basic algorithms through numerous examples provided in Emmanuelle Claeys' Master courses.

Afterward, I made an interactive prototype of the application using Balsamiq Wireframes. Finally, I began to actually program the application. The prototype proved to be a useful tool for iteratively

developing new functionalities while maintaining the source code clean and well-organized. Simultaneously, we held regular meetings on Zoom and Discord and used Google Drive for document and data sharing.

I.4 Advancement

In the application’s development, the *R* language was used for the implementation of algorithms and mathematical equations. On the other hand, *CSS* was employed to style graphical elements. The main sections are already available, allowing users to import datasets and visualize bees’ behavior through various graphics such as pie charts, bar charts, line charts, and boxplots. It shows the reward, the regret, and compares the results with popular Reinforcement Learning algorithms. Bees are also ranked according to their ability to maximize reward. In addition, two metrics provide intuition into the probabilistic strategy adopted by each individual and highlight their decision patterns. In future updates, users will be able to draw comparisons between the behaviors of different groups of bees, and new Reinforcement Learning algorithms will be added.

On the other side, I wrote reports and prepared oral presentations to meet both course requirements and the needs of researchers working at the CBI. After the internship, the project will be continued by other interns who will need to understand the key concepts, the data structures, and the different algorithms by virtue of clear documentation. Ultimately, we aim to deploy the application on the web, hosted on IRT’s servers.

In brief, the project is already at an advanced stage, although numerous improvements remain possible. It is therefore crucial to document the code, the decisions, and the organizational structure from the beginning, as this will facilitate future collaboration and ensure continuity.

I.5 Difficulties

Throughout this project, I withstand many difficulties. First and foremost, I had no prior experience with the *R* programming language and had to learn it on the heap. Subsequently, developing a web-style application is very time-consuming, especially when a single person is responsible for both front-end and back-end implementation. In other words, beyond the integration of reinforcement learning algorithms, considerable effort was required to design each interactive element and address visual bugs. Furthermore, a deep understanding of the biological experiment was crucial to appropriately process the collected data. Consequently, I spent a lot of time examining datasets and reviewing shared files on Google Drive. Lastly, the need to write reports and prepare presentations added another layer of complexity. Overall, the entire development phase, along with the organizational requirements typical of a genuine research project, proved to be particularly challenging.

I.6 Conclusion

In conclusion, this project aims to study honeybees’ decision-making processes using multi-armed bandit algorithms. Each bee is viewed as an agent trying to learn optimal behavior through interactions with its environment during a scientific experiment. The main goal was to develop a web-based application allowing researchers to import datasets and analyze bees’ strategies. A significant number of technical and organizational challenges emerged throughout the development. In fact, such a project requires careful planning, robust implementation, and detailed documentation to ensure continuity. However, a suitable tool was finally delivered.

A – Bibliography

- [1] Shiny - We Application Development Framework for R. Posit
<https://shiny.posit.co/>

- [2] Shinyapps.io - Website for Shiny Applications Hosting. Posit.
<https://www.shinyapps.io/>

- [3] Claey's Emmanuelle. banditWithR - R Package for Implementing Multi-Armed Bandit Algorithms.
<https://github.com/manuclaeys/banditWithR>

- [4] Airiau Rémi. BeesRLWithR - R Application for Analyzing Bee Decision-Making Process. Github.
<https://github.com/Ravemi987/BeesRLWithR>

- [5] Markov Decision Process. Wikipedia.
https://fr.wikipedia.org/wiki/Processus_de_d%C3%A9cision_markovien

- [6] Multi-Armed Bandit. Wikipedia.
https://fr.wikipedia.org/wiki/Probl%C3%A8me_du_bandit_mancho't

- [7] Multi-Armed Bandit : Data Science Concepts. Youtube.
<https://www.youtube.com/watch?v=e3L4VocZnnQ&t=5s>

- [8] Best Multi-Armed Bandit Strategy? (feat: UCB Method). Youtube.
<https://www.youtube.com/watch?v=FgmMK6RPU1c&t=3s>

- [9] Claey's Emmanuelle. Introduction to Multi-Armed Bandit. Canva.
https://www.canva.com/design/DAF-LSEeBw/VcExXyiGMoudF-vL_4p4mA/edit?ui=eyJJEIjp7IlAiOnsiQiI6ZmFsc2V9fX0